Skills Summary

Basic Matrix Operations

Use this reference while completing your worksheet or when studying.

Size first, always. A matrix with m rows and n columns is $m \times n$. Addition and subtraction need *identical* sizes; a product AB needs the columns of A to equal the rows of B, and then AB is (rows of A) $\times$ (columns of B).

Entry by entry: $(A \pm B)_{ij} = a_{ij} \pm b_{ij}$ and $(kA)_{ij} = k\,a_{ij}$.

Row meets column: $(AB)_{ij}$ is row i of A paired term by term with column j of B, then added. Order matters — AB and BA are usually different, and one may not even exist.

Determinant (2×2): $ad - bc$ for $\left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$. **Equal matrices** have equal corresponding entries.

1. Adding, subtracting and scaling

Check the sizes match, then work position by position. A scalar reaches *every* entry — the most common slip is multiplying only the first row. Nothing here mixes entries from different positions.

Example: Find $A + B$ for $A = \left[\begin{smallmatrix} 2 & -1 \\ 4 & 3 \end{smallmatrix}\right]$, $B = \left[\begin{smallmatrix} 5 & 2 \\ -3 & 1 \end{smallmatrix}\right]$.
Step 1. Both matrices are 2×2, so the sum exists and every entry of the answer comes from the matching pair of entries — no row-column pairing is involved.

Step 2. $2 + 5 = 7$, $-1 + 2 = 1$, $4 + (-3) = 1$, $3 + 1 = 4$. $A + B = \begin{bmatrix} \mathbf{7} & \mathbf{1} \\ \mathbf{1} & \mathbf{4} \end{bmatrix}$

Try it: Find $3A$ for $A = \left[\begin{smallmatrix} 2 & -5 \\ 1 & 4 \end{smallmatrix}\right]$.

check: $\begin{bmatrix} \mathbf{6} & \mathbf{-15} \\ \mathbf{3} & \mathbf{12} \end{bmatrix}$

2. Multiplying: row meets column

Slide row i of the left matrix down column j of the right one, multiplying the pairs you meet and adding the results. That single number is the entry in row i, column j. If the row and the column have different lengths, the product does not exist.

Example: Find AC for $A = \left[\begin{smallmatrix} 1 & 2 \\ 3 & 1 \end{smallmatrix}\right]$ and the column $C = \left[\begin{smallmatrix} 4 \\ 2 \end{smallmatrix}\right]$.
Step 1. A is 2×2 and C is 2×1: the inner dimensions agree, so the answer is a 2×1 column — one number per row of A.

Step 2. Row 1: $(1)(4) + (2)(2) = 8$. Row 2: $(3)(4) + (1)(2) = 14$. $AC = \begin{bmatrix} \mathbf{8} \\ \mathbf{14} \end{bmatrix}$

Try it: Find AC for $A = \left[\begin{smallmatrix} 2 & 3 \\ 1 & 4 \end{smallmatrix}\right]$ and $C = \left[\begin{smallmatrix} 1 \\ 3 \end{smallmatrix}\right]$.

check: $\begin{bmatrix} \mathbf{11} \\ \mathbf{13} \end{bmatrix}$

3. Determinants and matrix equations

The 2×2 determinant is main diagonal minus anti-diagonal: $ad - bc$. Subtract, never add. For a matrix equation, equal matrices means equal corresponding entries, so one matrix statement becomes one ordinary equation per position.

Example: Find the determinant of $\left[\begin{smallmatrix} 6 & 4 \\ 3 & 5 \end{smallmatrix}\right]$.
Step 1. A single number is wanted from a 2×2 matrix, and the determinant formula is the only operation that does that: main diagonal product minus the other one.
Step 2. Main diagonal $(6)(5) = 30$; anti-diagonal $(4)(3) = 12$; subtract. The determinant is **18**, and being nonzero it says the matrix is invertible.

Try it: Find x if $\left[\begin{smallmatrix} x-3 & 4 \\ 2 & 7 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 8 & 4 \\ 2 & 7 \end{smallmatrix}\right]$.

check: $x = \mathbf{11}$

(!) Watch out: matrix multiplication is not commutative — compute AB and BA separately, and expect different answers or a product that does not exist at all. When scaling, reach every entry. When taking a determinant, subtract the second product rather than adding it.